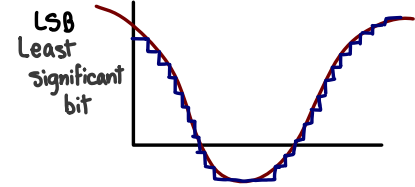


Analog ⇌ Digital Conversion

Goal for Analog / Digital Converter
Take V_{in} and convert to a stream of N-bit binary #s

$$V_{in} \in \{-V_{max}, V_{max}\} \Rightarrow 0-2^N-1$$



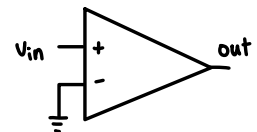
Quantization Error
= rounding error

ex.) $V_{in} \in \{-1V, 1V\} \Rightarrow N=4$
 $2V \Leftrightarrow 16$ binary values

$$\frac{2V}{2^4} = 0.125V \Leftrightarrow \text{LSB}$$

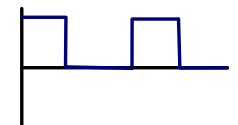
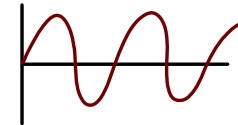
Error on a $\pm 10V$ signal
digitized @ 16 bit $\approx 300mV$

A/D for $N=16$ bit
use a comparator



$$V_{out} = \begin{cases} 1 & \text{for } V_{in} > 0 \\ 0 & \text{for } V_{in} < 0 \end{cases}$$

* continuous frequency of sine-wave



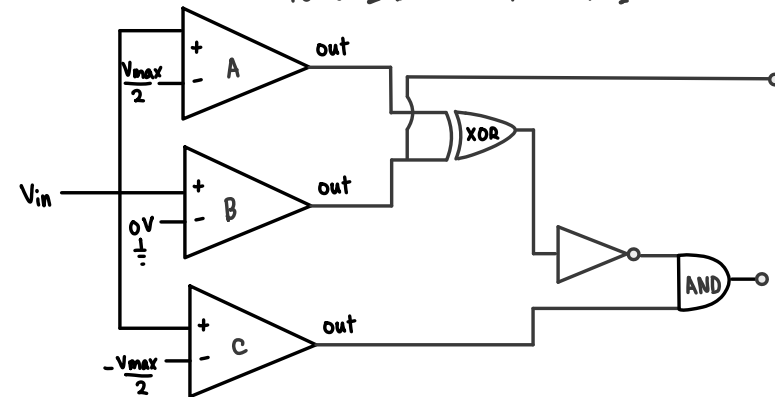
count the periods of square waves

$N=2$ Flash A/D converter

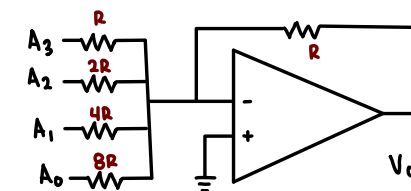
Design a Digital Logic Gate that corresponds V_{in} to a 2-bit binary # Q_1, Q_2

start w/ truth table:

A	B	C	Q_1	Q_2
0	0	0	0	0
0	0	1	0	1
0	1	1	0	1
1	1	1	1	1



D/A (DAC) - Summing Amplifier



$$A = A_3 A_2 A_1 A_0$$

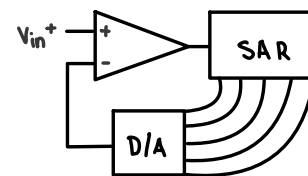
$$A \in 0..15$$

4-Bit binary

$$V_{out} = \frac{R}{R} A_3 + \frac{R}{2R} A_2 + \frac{R}{4R} A_1 + \frac{R}{8R} A_0$$

$$= A_3 + \frac{1}{2} A_2 + \frac{1}{4} A_1 + \frac{1}{8} A_0$$

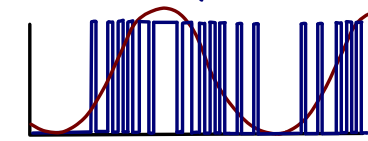
Successive Approximation Register



- Use a binary search to find each bit
assume $V_{in} \in (0, V_{max})$
- starting guess 1000 = $V_{max}/2$
- Look @ comparator output
 - If $V_{in} > \frac{V_{max}}{2}$ comp = 1, $Q \rightarrow 11000 \Leftrightarrow \frac{3V_{max}}{4}$
 - If $V_{in} < \frac{V_{max}}{2}$ comp = 0, $Q \rightarrow 01000 \Leftrightarrow \frac{1V_{max}}{4}$
- Look @ comparator output again & update guess of Q
w/ each iteration we nail down 1 bit

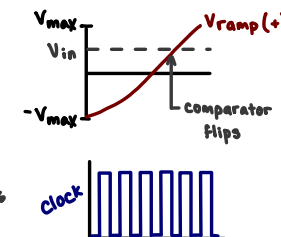
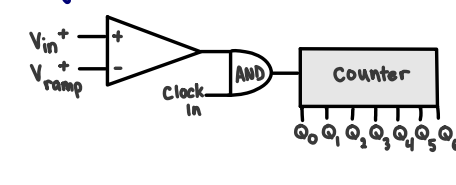
Fancier Idea (than PWM) for D/A Conversion

Sigma Converter ($\Delta\Sigma$)



- Vary Pulse Width + pulse rate
- Switch faster wherever possible (near 50% duty cycle)
- when you low pass filter, switching noise is reduced (smaller steps)

Ramp Converter
High Res, but slow



Over Sampling

- turn fast - low res
- turn slow - high res

* average a bunch of fast samples
4 avg $\Rightarrow$ 1 extra bit of res

only works if signal is a little noisy

$$V_{in} = 2.2V$$

$$0111 \Leftrightarrow 1.6V$$

$$1000 \Leftrightarrow 2.0V$$

$$1001 \Leftrightarrow 2.4V$$

But if you add $\pm 0.2V$ Noise

$\rightarrow$ Sometimes you get 1000
 $\rightarrow$ Sometimes you get 1001
 $\Rightarrow$ average 1000.1

with no noise,
will always measure the same thing
Averaging doesn't help

Digital Signal Processing (DSP)

Nyquist Theorem - Sampling Rate should be twice as fast as the signal frequency

Digital Samples - $X_1, X_2, X_3, \dots, X_n$

output - $Y_1, Y_2, Y_3, \dots, Y_n$

in class example Field Programmable Gate Array (FPGA)

$\rightarrow$ capable of taking analog signal in $\rightarrow$ converting to digital $\rightarrow$ transforming digital signal $\rightarrow$ returning analog signal
(emulated Low Pass / High Pass Filters, etc.)

In General:

$$Y_{n+1} = \sum_{i=1}^n a_i X_i + \sum_{i=1}^n b_i Y_i \text{ where } a_i \text{ or } b_i \text{ are weighting coefficients}$$

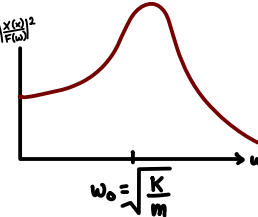
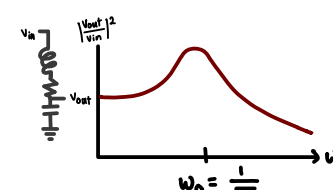
ex.)

$$R = 1k\Omega \quad T = 300^\circ K \quad \Delta f = 1MHz$$

$$\sqrt{V_r^2} = 4 \times 10^{-6} = 4\mu V$$

$$\Delta f = 100MHz \quad R = 10^6\Omega \quad \sqrt{V_r^2} \approx 1mV$$

H.O. look LRC circuits

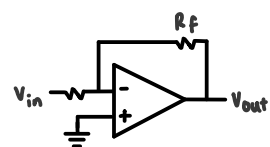


$$F(t) = -kx - m\Gamma \frac{dx}{dt} + F_{test}(+) + F_{feedback}$$

restoring force viscous damping $A \sin(\omega t)$

$$\text{ex: } -K_p x(t) - K_d \frac{dx}{dt}, \quad F(t) = -(k + K_p)x(t) - (m\Gamma + K_d) \frac{dx}{dt} + F_{test}$$

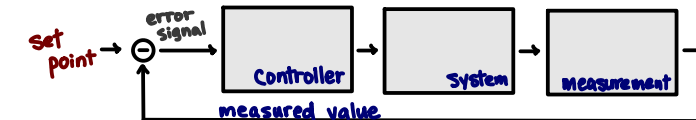
Feedback Control - use negative feedback to stabilize stuff (like signals)



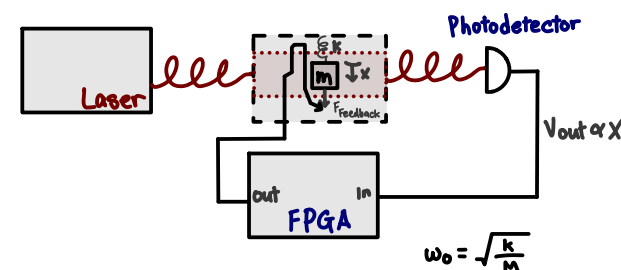
* use positive feedback to make things oscillate

ex: relaxation oscillator

In General:



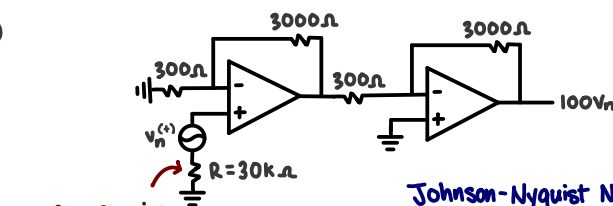
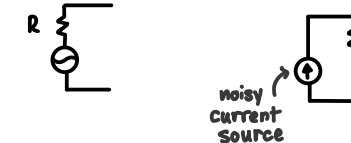
example:



Thermal Noise in Resistors @ Room Temperature, noise is random (and @ all frequencies)

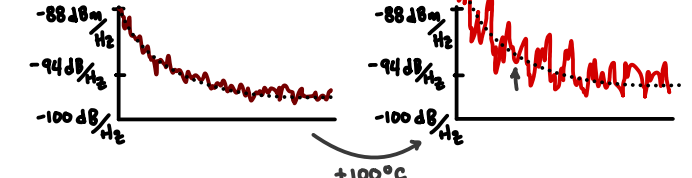
$$\bar{V}_n^2 = 4k_B T \cdot R \Delta f$$

Temp
measurement frequency
 $1.38 \times 10^{-23} J/K$



measure noise in this resistor

Johnson-Nyquist Noise



* increases V_{npp} and causes vertical shift